

Forecasting Security Incidents Using Geopolitical Data

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Abstract. Predicting security incidents is a profound task critical for informing proactive defensive measures and cyber-insurance policies. Prior work tackling this problem has mainly relied on structured, manually defined features based on network measurements (e.g., protocol misconfigurations). While demonstrating promising performance, these approaches require manual feature engineering (hence, may be challenging to maintain), and may fail to capture aspects related to adversaries’ motives. To fill this gap, our work leverages geopolitical data mined from public sources to forecast security incidents. Specifically, our approach relies on news articles and transcribed podcasts that are fed to large language models (LLMs) to automatically produce rich representations. The representations are then fed to a classifier trained to forecast future incidents based on historical ones. Our evaluation with a large incidents dataset (>15,700 records) demonstrates substantial accuracy (71.8% ROC AUC) with geopolitical data alone. Notably, combining geopolitical data and network measurements outperforms the state-of-the-art technique based on network features alone (82.0% vs. 77.5% ROC AUC). Our analysis also helps shed light on when geopolitical data is most helpful and the data sources that are most useful for accurate forecasting.

Keywords: Security analytics · Incident prediction · LLMs

1 Introduction

The increasing frequency and impact of cybersecurity incidents have motivated growing interest in forecasting potential incidents before they occur. Predictive cyber-risk modeling aims to identify organizations that are likely to suffer an incident within the near future [35,53], allowing for timely mitigation and resource allocation. Among others, such forecasts can help inform proactive defenses, for instance, by enabling experts to prioritize vulnerability patching—a core tenet of risk-based vulnerability management [49]. Beyond operational security, cybersecurity incident forecasting can also play a critical role in guiding cyber-insurance policies and risk assessment [47].

Traditional methods for forecasting cybersecurity incidents primarily rely on structured organizational and network-based features, such as untrusted certificates, DNS port randomization, exposed services, or vulnerability metrics [35,52,53].

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A notable example is the work of Liu et al. [35], which demonstrated that combining public vulnerability and (mis)configuration data can yield meaningful predictive signals for future attacks.

Still, although such structured data provide an important foundation, they capture only part of the picture. In practice, much of the contextual information surrounding an organization’s cyber posture—such as news coverage, partnerships, acquisitions, public exposure, or even mentions of past security issues—appears in unstructured text sources. Moreover, while public network indicators may capture the technical feasibility of attacks, they may not be able to capture attacker motives, as, for example, geopolitical data and news articles may capture [21]. Here, advances in large language models (LLMs) [12,57] provide new opportunities to extract latent patterns from such textual data at scale, potentially revealing precursors to breaches that are not reflected in technical telemetry alone.

In this work, we propose a framework that leverages geopolitical data and can further integrate unstructured text from articles with structured network features to predict future cybersecurity incidents. Using an LLM-based text encoder (§5), we transform news articles (e.g., ones mentioning target organizations) into dense semantic representations (i.e., vectors) that represent the surrounding context. These representations can then be used alone or combined with network features in a joint classification models for forecasting incidents. Our results indicate that our proposed approach based on LLMs using geopolitical data leads to forecasts with competitive accuracy in comparison to previous methods based on network indicators. Moreover, the incorporation (i.e., fusion) of textual representations and network features (§5.2) significantly improves prediction performance compared to network-only features, highlighting the complementary value of geopolitical data and technical indicators.

Our contributions are fourfold:

- We enrich a popular cybersecurity-incident dataset covering >15,700 incidents from the past 12 years [19] (§4.1), improving the accuracy of the reported incident dates, thus ensuring reliable model training and assessment.
- We collect and pre-process large-scale geopolitical dataset, consisting of news articles and podcasts, spanning over a decade, and reaching up to 2.5 million monthly records (§4.2). Moreover, we gather network features to reproduce seminal work by Liu et al. on predicting data breaches [35] and augment the input of our forecasting system (§4.4).
- We design an LLM-based system that ingests relevant geopolitical data to produce rich representations for forecasting security incidents (§5). Crucially, our system is extensible, enabling fusion between geopolitical data and network features to improve forecasts’ accuracy.
- We extensively evaluate our approach (§7), demonstrating the viability of leveraging geopolitical data for forecasting security incidents, and advancing the state-of-the-art by fusing geopolitical and network data for more accurate predictions. Furthermore, we uncover the settings where geopolitical data is most useful (e.g., predicting incidents motivated by protests), and charac-

terize the types of inputs that enable predictions (e.g., popular domains are a useful data source for predictions).

The remainder of this paper is organized as follows: We next review related work on forecasting events in cybersecurity and other domains (§2) before laying out our threat model (§3). Then, we describe our data-collection apparatus (§4), our forecasting system’s design (§5), and the limitations of our work (§6). Finally, we wrap the paper with the evaluation results (§7).

2 Related Work

In this section we present prior work on forecasting future events in computer security (§2.1) and other domains (§2.2).

2.1 Forecasting Security Incidents

Prior predictive security analytics work has primarily relied on system and behavioral measurements (e.g., [35,50,52,53]) and self-reported data (e.g., [52]) to collect observations and enable predictions. For instance, Soska and Christin collected information about content-management system versions and installed plugins to forecast whether web servers will be compromised within a year from data collection [53]. As another example, Sharif et al. gathered Internet browsing data (e.g., website categories and amount of bytes downloaded or uploaded), both historical and within browsing sessions, as well as users’ responses to standard questionnaires to predict impending exposure to malicious content on the fly, as users are browsing [52].

Most related to our work is the work of Liu et al. [35]. Their work aimed to forecast organizations’ security breaches by leveraging externally observable properties of the organizations’ networks. The researchers engineered and compiled different features falling into two main categories: mismanagement symptoms (such as misconfigured DNS or BGP records) and malicious activity time series (including spam, phishing, and scanning activity). Subsequently, using a machine-learning model, they demonstrated high prediction accuracy.

The features used for prediction in prior work (e.g., [35,50,52,53]) were mostly hand crafted by the researchers, requiring a laborious feature engineering and domain knowledge. Moreover, because humans may fail to discover specific correlations in the data, they may exclude important features conducive for accurate predictions. Not to mention that the manual work required for developing features may preclude timely updates to the system as new attacks emerge.

Differently from prior work, our work strives to leverage geopolitical news to forecast security incidents. Geopolitical data complement observations used in older systems in that they encode information describing attacker motives. Since motives play a central role in attacks [21], we expect that our work would inform analytics systems producing high-quality forecasts. As geopolitical information is usually available in the form of natural language, we build upon adequate

Year	Cite	Events	Time	Features	Model
2024	[23]	Paper cites	Days–years	Cite sequence	Attention nets
	[58]	World events (news)	Months–years	Articles (text)	Transformers
2022	[20]	Conflicts and military actions	Days	Articles (text)	LSTM
	[63]	World events (news)	Months–years	Articles (text)	Transformers
2021	[46]	Disease spread	Weeks	Time series and manual features	RNN
	[48]	Disease spread	Days	Spatio-temporal series	Diffusion
	[61]	Civil unrest and disease spread	Days	Multiple manual features	Custom deep model
2020	[14]	Traffic prediction	Minutes–hours	Traffic-speed time series	GNN
	[37]	Fluid resistance (physics)	Seconds	Physical system’s state	CNN
2019	[7]	Influenza spread	Months	Season and historical data	Deep clustering
	[22]	Social media user actions	Days	Historical actions and network	GNN
	[38]	System/sensor state	Unspecified	Sensor-state time series	RNNs

Table 1: A summary of academic papers studying the problem of forecasting events unrelated to computer security. For each paper, we report the event types forecasted; the time scales for the forecasts; the independent variables (i.e., features), including modality; and the predictive model’s type.

techniques to extract features necessary for prediction. Namely, toward addressing the shortcomings of prior work that relied on manual feature engineering, we incorporate automatic feature-extraction methods through LLMs [57].

2.2 Forecasting Events Unrelated to Security

A significant body of work studies the problem of forecasting events in domains other than computer security, ranging from science [23,37] to health [7,13,45,46,48], from different world events (e.g., civil unrest) [20,31,58,61,63] to social networks [22,62], and covering other domains as well [8,14,24,27,29,30,38]. Table 1 summarizes related work that appeared in peer-reviewed journals and archival conference proceedings within the past five years. As demonstrated from the Table 1, past work in domains other than computer security has primarily considered specific time scales for predictions (e.g., months) while offering little flexibility for changing these scales; mainly relied on manually crafted time-series and historical features; and mostly used dated (pre-transformer architecture) models. In contrast, our work, besides tackling predictions in computer security, leverages automatically extracted features from text; employs state-of-the-art transformer-based models; and enables forecasts at varying times-scales.

3 Threat Model and Problem Formulation

Our goal is to predict whether an organization will experience a cybersecurity incident within a given, pre-defined time window into the future. To this end, we parameterize the notion of cybersecurity incidents.

Definition 1. *τ -cyber incident:* Consider an organization *org* at time *t*. The organization is deemed to suffer from a τ -cyber incident if such incident is reported within time $\hat{t}$ such that $t \leq \hat{t} \leq t + \tau$.

With this definition, our formulation follows prior work on organizational data-breach forecasting [35], where forecasting is treated as a binary prediction problem. Formally, we define the cybersecurity-incident forecasting problem as follows:

Definition 2. *Cybersecurity-incident forecasting: For a given organization org at time t , the forecasting function F receives observations o_{org} and outputs a binary prediction $F(o_{org}) \in \{0, 1\}$ for whether a τ -cyber incident will occur.*

In practice, the output of $F(o_{org})$ may be a real number (e.g., a probability estimate $\in [0, 1]$) that would be binarized according to some threshold. We say that the F 's forecast is accurate if it outputs 1 (resp. 0) and an actual incident occurs (resp. does not occur) within $[t, t + \tau]$. In our research, we primarily consider a prediction window of $\tau=30$ days. This duration is consistent with prior work [35] and provides a window of opportunity for applying proactive defensive measures.

Cybersecurity Incidents. We aim to predict major cybersecurity incidents, akin to those reported in public media and recorded in cybersecurity-incident databases [19,54]. Such cyber incidents include, but are not limited to, denial of service attacks, data exfiltration, ransomware, or even physical attacks (e.g., manipulation of physical controllers or power grids) [19,54]. Note that, in certain jurisdictions and under specific conditions, regulations may obligate organizations to report security incidents that they suffer from. For instance, reporting is typically required for incidents that involve a high risk to the rights of individuals (e.g., European law requires notification of personal data loss within 72 hours of discovery), or those that have a material impact on public safety or financial markets (e.g., US regulations require reporting substantial cyber incidents within 72 hours and any ransomware payment within 24 hours of payment [9]). Such regulations help make our data more complete [19,54], thus enabling more reliable evaluation of forecasts' accuracy.

Observations (I.e., F 's Inputs, o_{org}). Our predictive model, F , operates solely on publicly available information, primarily derived from public media. It does not assume access to internal telemetry or attacker-specific data. For each organization, the forecasting model F receives contextual inputs composed of several information sources to enable predictions. As explained in §2.1, prior work leveraged technical indicators, such as DNS or BGP misconfigurations and appearance of organizational devices on blacklists, to forecast incidents [35]. In contrast, we focus on organization-related news collected from the written press and transcribed podcasts segments, forming geopolitical indicators at the level of the organization, its sector, or its country, to inform predictions. The geopolitical observations are collected from a historical time window $\in [t - h, t]$, and are later fed into an LLM (as part of F) to extract numeric features that are then mapped to predictions. For best performance, h can be tuned so that it is large enough to capture relevant information, but small enough so that the information is useful for predictions in the given time horizon $[t, t + \tau]$.

4 Data Collection and Processing

To train and evaluate forecasting models, we need both incident and observation data. The incident data serve as the dependent variables that the model seeks to predict, whereas the observation data serve as the independent variables. In both cases, as previously explained (§3), we rely on publicly available data. Doing so helps in maximizing reproducibility and transparency, and in obtaining broad, vendor-agnostic coverage without privileged access or intrusive measurement. For cybersecurity incidents (§4.1), we use public repositories recording security incidents discussed in the press. For observations, we primarily use geopolitical data from news articles and podcasts (§4.2), and consider augmentations with summary reports of global events (§4.3) and network measurements (§4.4).

4.1 Cybersecurity Incident Data

We use the Center for International and Security Studies at Maryland dataset (CISSM) [19] as the primary source of incident data. CISSM offers longitudinal, organization-level records of publicly reported cybersecurity incidents (2014 onward) curated under a consistent taxonomy and with public provenance (i.e., with references to public articles about incidents), making it well-suited for training and evaluating organization-level forecasting models. To date, CISSM contains >15,700 records about past incidents. Each record includes the organization name, event type, and a reported incident date, and, when available, information about the actor’s type and motivation, and the organization’s sector.

A practical caveat of CISSM, however, is that some records inherit the publication dates of news articles discussing incidents in lieu of the (earlier) incident dates stated in the articles’ body. To address this limitation, we used the Cyber Security Incident Database (CSIDB) [54] as a corrective layer over CISSM, as we found it mostly agrees with CISSM on incident dates, but is more accurate when it reports different (usually earlier) dates. Operationally, we matched CISSM incidents and with corresponding incidents in CSIDB via the cited source article. When a match was found, we replaced the CISSM timestamp with the incident date reported by CSIDB. A manual audit of records and their corresponding articles, showed that this process increased the accuracy of the incident date field from 77.1% to 91.7%.

4.2 Geopolitical Data Collection Pipeline

High-quality and high-coverage news data is essential for the forecasting model we use. Thus, we prioritized building an accurate and efficient data-ingestion pipeline based on large-scale data from reliable sources.

News Articles We collect our raw data from the CC-NEWS dataset, a continuously updated archive of global news extracted from Common Crawl [15]. Each monthly snapshot comprises of articles’ text (specifically, HTML pages) scraped from news sources across the web. These snapshots span multiple years, from

January 2016 onward, and offer a rich, multilingual resource for temporal and geopolitical analysis. Monthly volumes vary over time, peaking at approximately 2.5M monthly articles. We observed that, in every year, the maximal percentage of any single source is just below 4% of that year’s articles, indicating that the corpus remains diverse across years.

Podcasts To broaden the scope of organization and cyber-related media coverage, we collected a dataset of podcasts designed to capture both technical nuances and broader contextual signals. We selected 16 distinct sources categorized into two groups: (1) *Domain-specific cybersecurity outlets* (e.g., Darknet Diaries and The CyberWire) to provide granular details on vulnerabilities, incident forensics, and threat actor TTPs; and (2) *General news and geopolitical sources* (e.g., The Wall Street Journal, BBC, and NPR) to capture the economic instability and political friction that often precipitate major cyberattack campaigns. This variety ensures coverage of distinct layers of cyber and geopolitical events, extending from broad economic and political discussions to the reporting of emerging exploits and cyberattacks.

While we collected fewer podcast episodes than news articles, our podcast collection remains substantial in absolute terms, spanning diverse sources over a significant timeframe. In total, we collected over 30,000 podcast episodes, amounting to more than 14,000 hours of audio, each with a varied number of episodes, where the duration of each podcast varies from a just a few minutes up to nearly 200 minutes. These episodes span multiple years, going back as far as 2016 and reaching recent days in 2025.

Data Ingestion and Pre-Processing To efficiently process and extract useful content from new articles’ HTML files, we utilized the popular DataTrove framework [41], which implements pipelines for distributed processing of HTML files. Specifically, we used DataTrove’s boilerplate removal capabilities (based on Trafilatura [11]) to extract news articles’ text while removing irrelevant content from the raw HTML files. Moreover, using DataTrove’s language detector, we retained only articles written in English, as English is best supported by the state-of-the-art LLMs that our forecasting system utilizes. Last, for each article, we extracted the publication date via DataTrove. When unavailable, we defaulted to the CC-News’s timestamp otherwise. The resulting records were then inserted into a relational database to facilitate faster querying and linkage with other datasets.

Furthermore, we transcribed each podcast episode to enable subsequent analysis using LLMs. Specifically, we employed the open-source Whisper model [42] for transcription, finding it to be as accurate as commercial alternatives. We cleaned and segmented transcripts into smaller chunks, with a maximum length determined by an embedding model’s context limit (see below). Each segment was then indexed individually to enable later retrieval. We also added the transcripts to the relational database, complementing it with data and discussions that may be absent from traditional reporting (i.e., news articles).

Finally, all articles collected from CC-NEWS and the podcast transcriptions were encoded using a pretrained text embedding model to enable semantic

search. Specifically, the embeddings produced by the model can be used to retrieve text relevant to specific search queries, subsequently augmenting the input we feed into our LLM (§5). We selected `Arctic-embed-m-v2.0` [59] as our embedding model because it supports long context windows (up to 8,192 tokens), is relatively lightweight, and achieves strong retrieval results [36]. To support temporal retrieval, we stored the embeddings in a FAISS vector database [16] partitioned by month.

4.3 GDELT Contextual Data

Besides raw geopolitical news data, we sought to incorporate features to improve forecasts’ accuracy (§5.2). As one source of features, we used the Global Database of Events, Language, and Tone (GDELT) [33]—a large-scale open-source dataset that tracks worldwide news media, encoding information about events, entities, and tone from millions of sources in near real time. In lieu of using the raw data, we queried the hosted version of GDELT to extract specific aggregated indicators for each country and time period. Specifically, we extracted features related to media attention and volume (total mentions and total events), sentiment and stability (average tone and Goldstein scale), and specific geopolitical event types (material conflict events, protest events, and cooperation events). To mitigate daily volatility and capture sustained trends, we processed these features by calculating the average over a rolling window of $k=30$ months and then normalizing the current month’s value relative to this baseline.

4.4 Network Indicators

We collected network mismanagement and malicious-activity data, per Liu et al. [35], to construct a baseline forecasting approach [35] and to augment geopolitical news with additional indicators for better forecasting (§5.2). Specifically, we collected two complementary data types that describe organizations’ externally exposed posture: (1) *mismanagement symptoms* (static configuration features) that reflect longer-lived weaknesses in deployed infrastructure and policy, and (2) *daily malicious-activity signals* (time series).

Mismanagement Symptoms We collected four static, mismanagement-symptom features used by Liu et al. [35]. These features capture information about: *Open Recursive Resolvers*, identified via a December 2024 *Censys* [17] DNS snapshot (finding 4.8 million open resolvers); *insufficient DNS source port randomization*, where we measured the standard deviation of UDP source ports used by resolvers in response to 10 queries, establishing a criterion to distinguish proper from poor randomization; *untrusted HTTPS Certificates*, where we scanned public HTTPS endpoints from the December 2024 *Censys* snapshot with *ZGrab* [17,55], observing that 11.7 million IPs (>40%) presented untrusted certificates; *BGP misconfigurations (short-lived routes)*, where we analyzed BGP updates from 12 Route Views listeners over the December 2024 period and tracked the lifetime of newly announced routes [6,35]. Following prior work [60], we treat routes that persist

for less than 24 hours as misconfigurations, identifying more than 1.1 million unique short-lived prefixes. We clarify that we excluded misconfiguration features related to *open SMTP mail relays* used in prior work [35], as we found these to be extremely rare (only ~ 1 in 10,000 SMTP relays are open).

These collected features are not directly linked to specific vulnerabilities. Instead, the persistent presence of these technical misconfigurations in an organization’s network serves as an indicator of poor security hygiene and highlights shortcomings in the policies or controls designed for detecting and remediating failures. These shortcomings, as shown in prior work [35], consequently increases the potential risk of successful cyberattacks.

Malicious Activities Prior work [35] relied on a small, hand-selected set of eleven public blacklists. Nearly a decade later, at least four of these lists are no longer maintained. Therefore, we opted to use the Blacklist Aggregator (BLAG) [43] as the source of information about malicious activities. As part of its operation, BLAG continuously collects and archives the raw daily outputs of a large set of public blacklists (157 sources, including previously three used by Liu et al. [35]) covering (1) spam and abuse; (2) malware domains and hosts; and (3) scanning or attack activity. These blacklists date back to January 2019 and are continuously updated. We consume BLAG’s daily snapshots directly. In practice, a single day typically contains more than 1 million IP addresses. In our pipeline, when analyzing a certain organization in a given time period, we assess whether any of the organization’s IP addresses appear on the BLAG lists.

Data Pre-processing and Organization Mapping While our forecasting targets are organizations, the network data is linked to specific IP addresses of prefixes. To bridge this gap, inspired by Liu et al. [35], we mapped global IP addresses or address ranges to organizations using Regional Internet Registry (RIR) data from ARIN [3], RIPE [1], APNIC [2], AFRINIC [5] (LACNIC [4] was excluded due to logistical constraints). The data from these registries served two goals: (1) attributing IP addresses to victim organizations that appeared in CISSM; and (2) identifying non-victim organizations that were later used in our experiments (§7.1).

Due to the sheer size of our incident dataset compared to prior work’s [35] ($>10\times$ more incidents), manual mapping between organizations and RIR subnets was infeasible. Thus, we automated the mapping process by (1) standardizing organizations’ names in RIRs, as they tend to follow diverse naming conventions; (2) clustering RIR subnets according to the semantic similarity between the standardized names; and (3) mapping organizations from CISSM that suffered from security incidents to subnets according to semantic similarity between the organizations’ names. We manually assessed a statistically representative subset of the data in each of these steps to ensure high accuracy. Due to space constraints, the details are omitted and will be included in an extended version of this work.

5 Forecasting System

To enable cybersecurity incident prediction via geopolitical data, it is necessary to extract appropriate representations that can be fed into a predictive model. One of the most powerful means for extracting such representation from data is via LLMs [28,44]. Specifically, starting with pre-trained LLMs, their knowledge about a certain field is typically extended using a given corpus [34,18]. Subsequently, the representation they extract based on this extended knowledge enable predictions. In this section, we describe how we forecast cybersecurity incidents based on textual geopolitical data alone, and how we fuse geopolitical data with additional information to enhance prediction accuracy.

5.1 Forecasts Based on Textual Geopolitical Data

A popular paradigm for extending the knowledge of LLMs is through retrieval [34]. Here, the LLM’s context is augmented with relevant knowledge retrieved from the corpus using a dedicated dense-retrieval model [26,34]. The augmented context is then fed to the model which produces a representation that encodes the injected knowledge alongside the remainder of data included in the prompt.

Fig. 1 provides an overview of how we leverage retrieval to construct our forecasting system. Given an organization for whom we aim to forecast attacks, we use a retrieval model to find relevant information from the recent history that can aid the prediction. For instance, we may retrieve information about recent cyber attacks against the organization, whether recent policies or economic developments affected the organization, whether cyber attacks targeted other organization in the said organization’s country, etc., under the hypothesis that such information is correlated with cyber incidents that may target the organization in the near future. Particularly, we perform the retrieval from our pre-processed corpus of news articles and transcribed podcasts (§4.2). We then aggregate the retrieved content and extend it with an organization-specific prompt (e.g., “Will ORG suffer from a cyber incident in the following 30 days?”), and feed the resulting text to the LLM. Subsequently, the LLM produces a representation that we later feed to a classification head trained to forecast attacks.

We emphasize the importance of temporal validity when training and testing the forecasting system [40]. Suppose that a forecasting model is trained at time t_{tr} . In this case, when training the model to forecast an incident (or lack thereof) against an organization at time t_1 , the incident must have occurred at time $\in [t_1, t_1 + \tau]$, where $t_1 + \tau \leq t_{tr}$. Moreover, the model should only be evaluated when forecasting incidents at times $\geq t_{tr}$ to avoid forecasting into the past.

Another common approach for extending the knowledge of LLMs is through fine-tuning on the original pre-training approach (e.g., next- or masked-token prediction) [18]. Specifically, instead of providing relevant content in the input context, fine-tuning trains the LLM on the knowledge corpus offline, and no retrieval is conducted during deployment for augmenting the LLMs. Besides being more computationally expensive than retrieval, prior work also found that

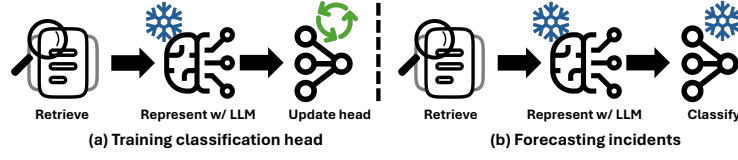


Fig. 1: An illustrative overview of how we leverage retrieval to forecast cybersecurity incidents. (a) At time t_{tr} , we train a classification head to predict security incidents at time $[t_1, t_1 + \tau]$, such that $t_1 + \tau \leq t_{tr}$ based on the representations produced by a frozen LLM. The LLM’s knowledge is augmented by feeding it relevant documents from time $\in [t_1 - h, t_1]$ that we retrieve using pre-specified queries. (b) At test time, to predict security incidents at time $\in [t_2, t_2 + \tau]$, such that $t_2 \geq t_{tr}$, we produce representations using the same frozen LLM, through feeding it relevant documents retrieved from time $\in [t_2 - h, t_2]$. The representations are then fed to a frozen classification head to estimate the likelihood of a cybersecurity incident.

fine-tuning often falls short in terms of performance [39,25]. Our experiments, too, showed favorable results with retrieval compared to fine-tuning.

5.2 Forecasts Based on Fusion

In addition to using textual geopolitical data alone to forecast security incidents, we also explored means to incorporate other indicators to enhance prediction accuracy. In particular, we studied means to integrate data from GDELT (§4.3) and the network measurements (§4.4) alongside the geopolitical data into forecasts. The former may complement our geopolitical data by capturing general trends that may not be reflected in the retrieved content fed to the LLM. The latter may capture information related to the technical feasibility of attacks that complement the geopolitical content that mainly captures attacker motivation.

We considered two fusion techniques, including input fusion and hybrid fusion [10]. In the input-fusion approach, we enrich the textual prompt with a natural language summary of the organization’s country-level geopolitical context. From GDELT’s diverse feature set, we selected *material conflict events* (e.g., physical clashes), *total geopolitical events*, and *total media mentions*. To effectively integrate these statistics, we applied a 30-month sliding window to normalize the values, comparing each month’s count against the average of the preceding 30 months. We then discretized the values into intensity levels (e.g., “below normal”, “unusually high”) and determined recent trends, mapping them to templates such as: “*Media attention toward the country was unusually high and continues to rise recently.*” We adopted this normalization strategy because our preliminary analysis indicated that the deviation from historical norms exhibited a strong correlation with the frequency of cyber incidents within the certain countries.

In hybrid fusion, we combined input fusion with intermediate, feature-level fusion. Specifically, we incorporated network features directly into the classification head while aggregating it with the representation produced by the LLM for the geopolitical and GDELT data. More specifically, we extract the LLM representation and concatenate it with the feature vector extracted from network data, per Liu et al. [35]. The resulting joint representation is then fed into the classification head.

6 Limitations

Several limitations should be taken into account when assessing our forecasting system and interpreting our results. First, some organizations may not report cybersecurity incidents [47] and certain incidents may not be captured by CISSM or CSIDB. This may stem, among others, from lacking policies mandating the reporting of cybersecurity incidents [51], or inherent biases in cybersecurity incident repositories. Consequently, the recall (i.e., true positive rates) of our system may be underestimated, while the false positive rates may be overestimated. Moreover, the training data we use may potentially be noisy, thus harming accuracy. In other words, more complete incident reporting is expected to lead to improved system performance. As all methods we compare are equally affected, we anticipate that our conclusions regarding their relative performance would remain intact.

Second, our manual assessment showed that merging from CSIDB and CISSM has improved incident dates’ accuracy. Nonetheless, certain inaccuracies remain, potentially harming the evaluation’s precision. It was infeasible to manually update the dates of thousands of incident records, and other automated approaches we tested (e.g., using conversational agents) did not improve dates’ accuracy. Still, we expect that making the dates more accurate (e.g., via improved collection methodology in CISSM or CSIDB) would also help make the evaluation results more precise.

Finally, while we did our best to precisely map organizations to network indicators (§4.4), some imprecisions persist. Prior work conducted manual mapping, which remained feasible with the relatively small dataset considered. In contrast, we consider a significantly larger dataset precluding manual mapping. However, a sample we manually checked showed that our mapping was highly precise. Thus, we expect that our evaluation results of models leveraging network-based indicators would remain mostly unchanged even with a more accurate mapping procedure.

7 Evaluation

7.1 Experimental Setup

Models As an LLM, we used LLaMA-2-7B [56] due to its relatively early cut-off date (September 2022) and the superior accuracy it enabled compared to

Table 2: The queries used to retrieve relevant geopolitical data to enable forecasts.

Q1: “Are ORG and similar companies at risk of cyber attacks?”
Q2: “What are major economic or political developments in ORGCOUNTRY that could affect businesses?”
Q3: “What regulations or policy changes have recently impacted ORG in ORGCOUNTRY?”
Q4: “Has ORG been involved in any controversies, leadership changes, or restructuring?”
Q5: “Are there known conflicts between ORGCOUNTRY and other nations that might influence cyber or economic targeting?”
Q6: “Has ORG received significant public or media attention recently that could attract adversarial interest?”

other models we tested (e.g., Longformer). We extracted representations from an intermediate layer (specifically, 23 of 32), as it led to better forecasts than later layers. Subsequently, we fed the representations to a multilayer perceptron (MLP) that we used as a classification head (three hidden layers with 256 neurons each followed by GELU activations) trained via binary cross-entropy.

Retrieval Configurations For retrieval, we used the queries listed in Table 2. These queries, as hypothesized earlier (§5.1), led to high forecast accuracy. For each query, when predicting incidents at time t , we extracted most relevant passages from up to $h=9$ months prior to t . As noted before (§4.2), we used the `Arctic-embed-m-v2.0` retrieval model for best performance.

Metrics We adopted the metrics used by Liu et al. [35] for evaluation. Specifically, we computed receiver operating characteristic (ROC) curves, which report true positive rates (TPRs) and false positive rates (FPRs) at different operating points. The higher are the TPR and the area under the curve (AUC) and the lower is the FPR, the better is the forecast accuracy.

Data Splits We derived the victim organizations and incident dates from our incident dataset (§4.1). Unless stated otherwise, for each victim, we set the prediction date to be within $\tau=30$ days before the actual incident (selected uniformly at random). We focused on this 30-day window to prioritize immediate tactical warning signals over long-term strategic forecasting. However, we also tested prediction up to $\tau=1$ year into the future.

For non-victims, to mitigate model reliance on spurious static features like the organizations’ names, we implemented a two-stage sampling strategy. Starting with non-victim organizations sampled from the Regional Internet Registry (RIR) records, we identified that a simple linear classifier based on 1-grams could distinguish victims from non-victims based solely on naming artifacts (ROC AUC >0.5). Therefore, we prioritized non-victim organizations RIR records that the 1-gram classifier (incorrectly) predicted as victims. Subsequently, we augmented the set of non-victims with organizations that suffered from incidents

at some point; however, we were careful to sample these organizations at least one year away from the reported incident date to account for potential reporting delays. Our sampling approach forced the model to ignore spurious characteristics, such as organization names, and instead focus on geopolitical changes that indicate potential security incidents.

Overall, we ran our evaluation on 16,954 records, with 7,120 pertaining to victim organizations (i.e., ones who suffered from an incident at a given time) and 9,834 belonging to non-victim organizations (i.e., ones who did not suffer from an incident at a given time). For better reliability, we split the data at random into five folds, with $\sim 80\%$ of samples used for training and the test samples selected from the remaining $\sim 20\%$, and averaged our results across folds. To ensure temporal validity, as discussed previously (§5.1), when testing predictions at time t , we avoided training our models on any data (dependent or independent variables) from a later time $> t$. Hence, we gradually trained models till particular target dates, and tested them on the $\tau=30$ days immediately following these dates. Moreover, our test samples were taken only from 2024 and 2025, after the cut-off date of the LLM we used.

Baseline As a baseline method to compare our models against, we implemented Liu et al.’s [35] forecasting model based on network indicators. Specifically, we implemented a Random Forests (RF) classifier that takes misconfiguration and malicious activity features as input and predicts the likelihood of attacks. Besides optimizing the hyperparameters of the RF model, we further improved the performance of the model using an MLP for prediction. The comparison with this baselines serves two goals. Primarily, it shows where our approach stands compared to the state-of-the-art cybersecurity incident forecasting method. Second, it helps assess how geopolitical data (mostly encoding adversary motives) compares to network indicators (mostly indicating organizations’ security hygiene) when predicting attacks.

7.2 Experimental Results

Geopolitical vs. Network Data Fig. 2a shows the forecasting accuracy of our LLM-based approach using geopolitical data and compares it to baselines relying on network features. As perhaps expected for short-term predictions ($\tau=30$ days), network indicators that capture misconfigurations and malicious activity from organizations’ networks achieve higher accuracy (74.4% ROC AUC for the original RF-based model, and 77.5% AUC after our improvement with an MLP). Nonetheless, our approach attains competitive performance, with 71.8% AUC, confirming the viability of geopolitical data that helps capture the adversary’s motivation as a source of information for predicting cybersecurity incidents.

Fusion Leads to Best Performance Notably, fusing geopolitical data and network indicators leads to the best forecasting accuracy (see Fig. 2b). Input fusion with GDELT features slightly pushes the performance of our system (from 71.8% to 72.5% AUC). However, the most pronounced improvement is achieved

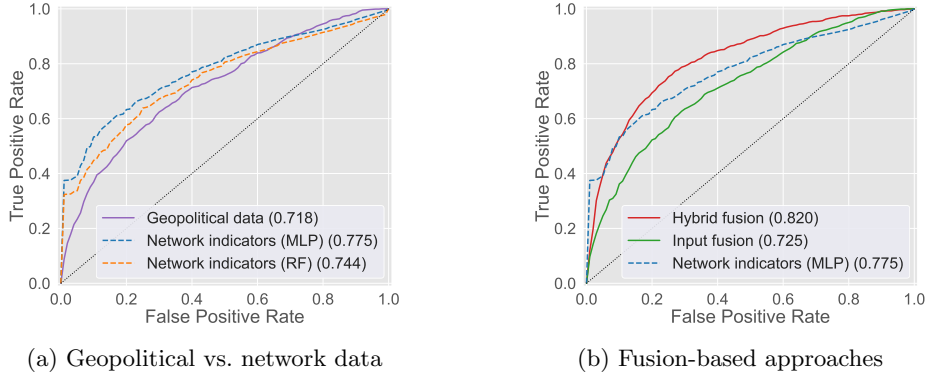


Fig. 2: ROC-curve comparisons: (a) presents the performance of our approach using geopolitical data alone against the baseline based on network indicators, while (b) shows the performance of fusion-based forecasting.

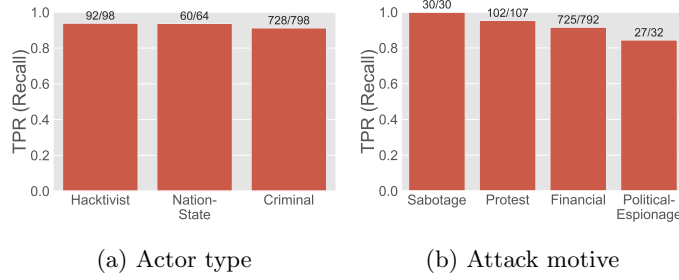


Fig. 3: Forecasting TPRs (at max F1 score) of our approach using geopolitical data alone for incidents involving different adversary (a) types (b) and motives.

via hybrid fusion, with GDELT features in the input, and network features in an intermediate phase, leading to $\sim 5\%$ higher ROC AUC compared to the baseline based on an MLP (82.0% vs. 77.5% AUC). This result shows that combining feature types that encode information about both adversary motives and organizations' security hygiene yields substantially better forecasts.

Predictions Into Longer Horizons Are Possible Using our data splits, we also evaluated predictions $\tau=1$ year into the future. In particular, we tested whether models trained until 2023 or 2024 can predict incidents that have occurred during these years. Fig. 4 presents the results. Here too, we find that the MLP based on network indicators alone attains strong results, outperforming the model fusing geopolitical and GDELT data at the input (77.3% vs.

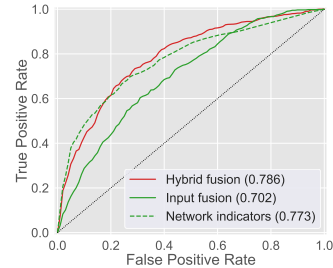


Fig. 4: Predicting incidents 1 year into the future.

70.2% AUC). However, hybrid fusion, combining both geopolitical and network data provides the most accurate forecasts (78.6% AUC).

When Is Geopolitical Data Most Helpful? To uncover when geopolitical data is most helpful, we analyzed how forecasting accuracy varies across adversary types, adversary motives, and organization sectors, as reported in CISSM. For each of these, we measured and reported the TPR of our model based on geopolitical data alone where the F1-score is maximal (Fig. 3). The results show intuitive changes across categories. For adversary types, we find that the prediction accuracy is higher for nation-states and hacktivists (that may be more driven by geopolitical events) than for criminals (who may be more opportunistic). Along the same lines, for adversary motives, we find that attacks motivated by sabotage and protest are more accurately predicted than other attacks (e.g., financially motivated ones). Finally, there is less variance across sectors, yet, we find that incidents against organizations in healthcare and education are predicted with slightly higher accuracy than incidents targeting other industries.

What Makes Our Approach Work? To better understand what makes our approach succeed, we analyzed how the context affects prediction accuracy when using geopolitical data alone. Primarily, we ablated the queries used to for retrieving documents to form the context to our model (Table 2) and measured how excluding each query affects the ROC AUC. We found that Q1 (retrieving text related to the cyber risk of organizations) and Q3 (finding documents on country-level regulations that may impact organizations) were most critical—excluding led to a 4.2% AUC drop.

Furthermore, we explored what sources of geopolitical news were played a central role in forming the context fed to our LLM (Fig. 5). We found that popular websites ranked part of the top 10,000 Tranco¹ domains [32] consistently contributed almost 20–40% of the retrieved articles selected to build the LLM’s ingested context. Interestingly, this is significantly higher than the overall rate of such articles in the general corpus we collected: roughly only 1–5% of articles were from websites in the top 10,000. In other words, articles from popular domains (e.g., Yahoo News) played a vital role in enabling predictions despite being relatively rare. This indicates that defenders may not need to rely on less reputable sources to forecast potential cyber incidents.

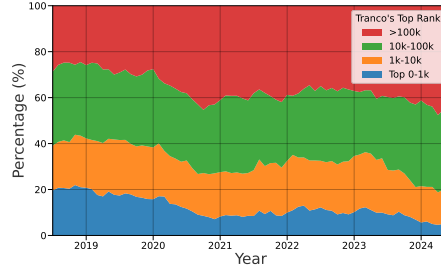


Fig. 5: The fraction of retrieved documents categorized by their Tranco rank.

¹ List available at <https://tranco-list.eu/list/4N6KX> (accessed 17 Dec 2025).

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